

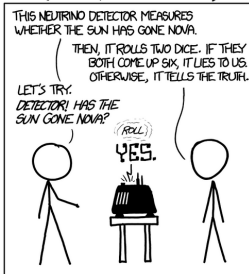
Econometrics I

Lecture 3a: Inference and Standard Errors

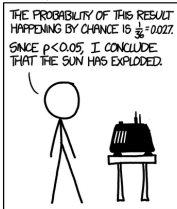
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DID THE SUN JUST EXPLODE? (IT'S NIGHT, SO WE'RE NOT SURE)



FREQUENTIST STATISTICIAN:



BAYESIAN STATISTICIAN:



Before the formulas: what are we uncertain about?

- ▶ Every inference question starts with an **estimand**: the number you would compute with infinite data. An elasticity. A treatment effect. The gender gap in log wages. (Goff's Appendix C calls this "defining the parameter of interest," and puts it *before* estimation.)
- ▶ A standard error quantifies exactly one kind of uncertainty: **sampling uncertainty** — you have n observations instead of all of them.
- ▶ It says *nothing* about the other ways to be wrong: omitted variables (last week), measurement error, the wrong functional form, the wrong estimand.
- ▶ Corollary you already simulated: the short regression in the Silly Example has a **tiny standard error around the wrong number**. Precision is not validity.

Today: how to get the sampling uncertainty right. Keep asking the other questions anyway.

Recap: Asymptotics for OLS and the Linear Model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Recall the three basic OLS assumptions

1. $\mathbb{E} [\varepsilon_i | x_i] = 0$
2. $(x_i, y_i), i = 1, \dots, n$, are i.i.d.
3. Large outliers are rare: $\mathbb{E} [y_i^4] < \infty$ and $\mathbb{E} [x_i^4] < \infty$.

- Unbiasedness means that, on average, an estimator $\hat{\theta}$ neither over- nor under-estimates the true θ :

$$\mathbb{E}[\hat{\theta}] - \theta = 0$$

- Consistency tells us that we approach the true θ as $n \rightarrow \infty$.

$$\hat{\theta} \xrightarrow{p} \theta$$

- Example: X_1 (the first observation) is unbiased but not consistent for the mean.
- Example: $\frac{n}{n-5} \bar{X}$ is consistent but biased for the mean.

- ▶ **Outliers** refer to observations that are “far away” from the rest of the data. They can be due to errors in the data. There is no standard formal definition.
- ▶ What to do? Greene: *“It is difficult to draw firm general conclusions... It remains likely that in very small samples, some caution and close scrutiny of the data are called for.”* I’d say that’s true even in large samples, but there isn’t a generally accepted way of quantifying what counts as appropriate “caution and close scrutiny.”

Removing Outliers?

- ▶ Removing extreme outliers (in x) from datasets is often considered good practice. But we should be mindful about why, as dropping observations creates the potential for manipulation.
- ▶ Sometimes extreme outliers are just errors, in which case they should almost certainly be dropped.
- ▶ Even if they aren't errors, they may reflect a different mode in the data generating process. They may require a different or more general model to account for them properly. Consider the justification of a linear model based on Taylor's theorem (local linear approximation). With such a justification for your modeling strategy, it would not make sense to include an outlier in x .
- ▶ It's important to be transparent about how dropped outliers affect results.

- One way to find influential observations is to calculate the **leverage** of each observation i . We begin with the hat matrix:

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

and consider the diagonal elements, which are labeled h_{ii}

$$h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$$

- This tells us how influential an observation is in our estimate of $\mathbf{b}_{OLS}$. Particularly important for $\{0, 1\}$ dummy variables with uneven groups.

Leave One Out Regression

- ▶ This is sometimes called the **Jackknife**
- ▶ Sometimes it is helpful to know what would happen if we omitted a single observation i
- ▶ Turns out we don't need to run n regressions

$$\begin{aligned}\mathbf{b}_{-i} &= (\mathbf{X}_{-i}'\mathbf{X}_{-i})^{-1}\mathbf{X}_{-i}'\mathbf{y}_{-i} \\ &= \mathbf{b}_{OLS} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\tilde{e}_i \quad \text{where } \tilde{e}_i = (1 - h_{ii})^{-1}e_i\end{aligned}$$

- ▶ $\tilde{e}_i$ has the interpretation of the LOO prediction error.
- ▶ high leverage observations move $\mathbf{b}_{OLS}$ a lot.

- ▶ A useful identity for linear algebra (scalar a , fixed matrix $\mathbf{A}$, random vector $\mathbf{z}$):

$$\mathbb{V}(a\mathbf{z}) = a^2\mathbb{V}(\mathbf{z})$$

$$\mathbb{V}(\mathbf{A}\mathbf{z}) = \mathbf{A}\mathbb{V}(\mathbf{z})\mathbf{A}'$$

- ▶ Since $\mathbf{b}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$,

$$\mathbb{V}(\mathbf{b}_{OLS} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbb{V}(\mathbf{y} \mid \mathbf{X}) \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}$$

- ▶ Recalling that $\mathbb{V}(\mathbf{y} \mid \mathbf{X}) = \mathbb{V}(\boldsymbol{\varepsilon} \mid \mathbf{X})$ (because $\mathbf{X}\boldsymbol{\beta}$ is fixed given $\mathbf{X}$),

$$\mathbb{V}(\mathbf{b}_{OLS} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbb{V}(\boldsymbol{\varepsilon} \mid \mathbf{X}) \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}$$

Start with the variance of the errors to form a **diagonal** matrix $\mathbf{D}$:

$$\mathbb{V}(\boldsymbol{\varepsilon}|\mathbf{X}) = \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}] = \mathbf{D}$$

$$\mathbf{D} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) = \begin{pmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{pmatrix}$$

- ▶ $\mathbf{D}$ is diagonal because, for $i \neq j$, $\mathbb{E}[\varepsilon_i \varepsilon_j | \mathbf{X}] = \mathbb{E}[\varepsilon_i | \mathbf{x}_i] \cdot \mathbb{E}[\varepsilon_j | \mathbf{x}_j] = 0$ (independence across observations)
- ▶ The diagonal elements of $\mathbf{D}$ are $\mathbb{E}[\varepsilon_i^2 | \mathbf{X}] = \mathbb{E}[\varepsilon_i^2 | \mathbf{x}_i] = \sigma_i^2$.
- ▶ In the **homoskedastic** case $\mathbf{D} = \sigma^2 \mathbf{I}_n$.

Variance of $\mathbf{b}_{OLS}$

Let $\tilde{\mathbf{D}} = \text{diag}(\varepsilon_1^2, \dots, \varepsilon_n^2)$. Then

$$\mathbf{D} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) = \mathbb{E}[\tilde{\mathbf{D}} \mid \mathbf{X}]$$

An (infeasible, since ε is unobserved) estimator of $\mathbf{v}_b = \mathbb{V}(\mathbf{b}_{OLS} \mid \mathbf{X})$ plugs in $\tilde{\mathbf{D}}$ for $\mathbf{D}$:

$$\hat{\mathbf{v}}_b^{\text{ideal}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\tilde{\mathbf{D}}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \varepsilon_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

The expectation shows us this estimator is unbiased:

$$\begin{aligned} \mathbb{E}[\hat{\mathbf{v}}_b^{\text{ideal}} \mid \mathbf{X}] &= (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \mathbb{E}[\varepsilon_i^2 \mid \mathbf{X}] \right) (\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \sigma_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\mathbf{D}\mathbf{X}) (\mathbf{X}'\mathbf{X})^{-1} = \mathbf{v}_b \end{aligned}$$

Heteroskedasticity Consistent (HC) Variance Estimates

Replace the unobserved ε_i^2 with squared OLS residuals e_i^2 :

$$\hat{\mathbf{v}}_{\mathbf{b}}^{HC0} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

$$\hat{\mathbf{v}}_{\mathbf{b}}^{HC1} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1} \cdot \left(\frac{n}{n-K} \right)$$

Could use leave-one-out residuals $\tilde{e}_i = (1 - h_{ii})^{-1} e_i$:

$$\hat{\mathbf{v}}_{\mathbf{b}}^{HC2} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-1} \mathbf{x}_i \mathbf{x}_i' e_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

$$\hat{\mathbf{v}}_{\mathbf{b}}^{HC3} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-2} \mathbf{x}_i \mathbf{x}_i' e_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

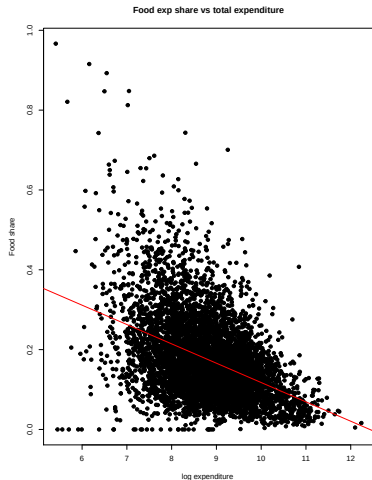
Heteroskedasticity Consistent (HC) Variance Estimates

- ▶ We know that $\hat{\mathbf{V}}_{\mathbf{b}}^{HC3} \geq \hat{\mathbf{V}}_{\mathbf{b}}^{HC2} \geq \hat{\mathbf{V}}_{\mathbf{b}}^{HC0}$ (in the positive-semidefinite sense) because $0 < 1 - h_{ii} \leq 1$.
- ▶ $HC3$ are the most **conservative** and also place the most weight on high-leverage observations.
- ▶ Stata uses $HC1$ as the default and it is what most people refer to when they say **robust standard errors**.
- ▶ These are often called White (1980) SE's or Eicker-Huber-White SE's.
- ▶ In small samples there is some evidence that $HC2$ has better **coverage** (what is that?)

Example: Engel Curves

- ▶ Engel curves refer to the relationship between a household's expenditure share on a good and income (or total expenditure).
- ▶ Engel curves for food are typically downward sloping – as total expenditure of a household increases, the proportion of its expenditure dedicated to food falls.
 - Expenditure on food still rises as total expenditure rises, but less than proportionally, so that food's expenditure *share* falls.

Food Engel Curves



- If we plotted total food expenditure (rather than the expenditure share), the heteroskedasticity would go in the other direction.

Data source: BLS Consumer Expenditure Survey data

- ▶ Recall that we defined the homoskedasticity assumption as:

$$\mathbb{V}(\boldsymbol{\varepsilon}|\mathbf{X}) = \sigma^2 \mathbf{I}$$

this assumption has two aspects:

1. The disturbance for each observation has the same variance
 2. Imposing zero correlation between disturbances for different observations
- ▶ The terminology can be misleading here, because what people refer to as “heteroskedasticity-robust” standard errors (the HC0–HC3 estimators above) are robust to violations of 1 but not 2.
 - ▶ We need to do a bit more to estimate standard errors in a way that is robust to correlated data.

- ▶ The baseline assumptions of the linear regression framework imply that the disturbances are uncorrelated across observations. There are many ways for this to be violated.
 - Example 1: we might have county-level data for a regression and be concerned that different counties within a given state have correlated disturbances because all counties are subject to the same (unobserved) state-level policies.
 - Example 2: time series data (asset prices), and we are worried that some unobserved factors within the disturbances are serially correlated
 - Example 3: county level data again, and we are worried about geographically correlated factors such as weather.

Different correlation patterns call for different estimators of $\mathbf{V}_{\mathbf{b}} = \mathbb{V}(\mathbf{b}_{OLS}|\mathbf{X})$. Some common alternatives to the no-correlation baseline:

1. Clustered standard errors, when there is correlation between observations within well-defined groups, but no correlation between observations in different groups.
2. Newey-West standard errors (and extensions) to deal with serial correlation in time series data.
3. Conley-Newey-West standard errors that allow for correlation in multiple dimensions (especially popular in the context of spatially explicit models).

What is Clustering?

Suppose we want to relax our i.i.d. assumption:

- ▶ Each observation i is a villager and each group g is a village
- ▶ Each observation i is a student and each group g is a class.
- ▶ Each observation t is a year and each entity i is a state.
- ▶ Each observation t is a week and each entity i is a shopper.

We might expect that $\mathbb{C}(\varepsilon_{ig}, \varepsilon_{jg}) \neq 0$ for $i \neq j$ in the same group $g \rightarrow$ independence is a bad assumption.

The groups (villages, classrooms, states) are independent of one another, but within each group we can allow for arbitrary correlation.

- ▶ If correlation is within an individual over time we call it **serial correlation** or **autocorrelation**
- ▶ Just like in time-series $\rightarrow$ we have fewer effective independent observations in our sample.
- ▶ Asymptotics now about the number of groups $G \rightarrow \infty$ not observations $n \rightarrow \infty$

- ▶ Suppose data are organized into distinct groups $g = 1, 2, \dots, G$. Let $g(i)$ be the group identity of observation i .
 - e.g., with county-level data, we have $g(\text{Manhattan}) = \text{NY}$.
- ▶ We assume $\mathbb{E} [\varepsilon_i \varepsilon_j | \mathbf{X}] = 0$ as long as $g(i) \neq g(j)$, and we do not restrict the correlation $\mathbb{E} [\varepsilon_i \varepsilon_j | \mathbf{X}]$ for observations within the same group.
- ▶ Intuition: the linear regression framework with no correlation in observations will overstate the precision of our estimates. If we add another observation within a cluster, and that observation is highly correlated with the other observations, it's not actually as good as adding another independent observation.

- Recall the sandwich formula for standard errors:

$$\mathbb{V}(\mathbf{b}_{OLS}) \approx n^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{V}[\mathbf{x}_i \varepsilon_i] \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1}.$$

$$\hat{\mathbb{V}}[\mathbf{x}_i \varepsilon_i] = n^{-1} \sum_i \mathbf{x}_i \mathbf{x}_i' e_i^2$$

- With clustering, the middle term also picks up within-group cross products:

$$\hat{\mathbb{V}}_{clu} = n^{-1} \sum_{i=1}^n \sum_{j=1}^n \mathbf{x}_i \mathbf{x}_j' e_i e_j \cdot \mathbb{I}[g(i) = g(j)]$$

where the indicator is 1 when i and j are in the same group and 0 otherwise. The $i = j$ terms are exactly the heteroskedasticity-robust sum above.

- ▶ The cluster-robust estimate of standard errors will be consistent as the number of groups gets large.
- ▶ Note that this estimator adds extra terms $\mathbf{x}_i \mathbf{x}_j' e_i e_j$ (covariance terms) to the estimate of variance, so it makes standard errors larger when $\mathbf{x}_i \varepsilon_i$ is positively correlated within groups — typically because *both* the regressor and the error are.
- ▶ Rough guide (Moulton): the variance is inflated by about $1 + (n_g - 1) \rho_x \rho_\varepsilon$, where ρ_x and ρ_ε are the within-group correlations of the regressor and the error. A regressor that varies only at the group level ($\rho_x = 1$) is the worst case; one that is independent within groups ($\rho_x \approx 0$) barely changes.
- ▶ Thus, if standard formulas are used in the presence of cluster-correlated disturbances and regressors, standard errors will be too small.
- ▶ Statistical software packages typically make it easy to compute cluster-robust errors.
- ▶ Clustering often makes a **huge** difference in standard errors.

Clustering Derivation

Begin by stacking up observations in each group, $\mathbf{y}_g = (y_{1g}, \dots, y_{n_gg})'$; we can write the model three ways:

$$y_{ig} = \mathbf{x}'_{ig}\boldsymbol{\beta} + \varepsilon_{ig}$$

$$\mathbf{y}_g = \mathbf{x}_g\boldsymbol{\beta} + \boldsymbol{\varepsilon}_g$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

All of these give the same OLS estimator:

$$\mathbf{b}_{OLS} = \left(\sum_{g=1}^G \sum_{i=1}^{n_g} \mathbf{x}_{ig}\mathbf{x}'_{ig} \right)^{-1} \left(\sum_{g=1}^G \sum_{i=1}^{n_g} \mathbf{x}_{ig}y_{ig} \right)$$

$$\mathbf{b}_{OLS} = \left(\sum_{g=1}^G \mathbf{x}'_g\mathbf{x}_g \right)^{-1} \left(\sum_{g=1}^G \mathbf{x}'_g\mathbf{y}_g \right)$$

$$\mathbf{b}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\mathbf{y})$$

Clustering Derivation (Continued)

The error terms have covariance within each cluster g as:

$$\Sigma_g = \mathbb{E} [\boldsymbol{\varepsilon}_g \boldsymbol{\varepsilon}_g' \mid \mathbf{x}_g]$$

In order to calculate $\mathbf{V}_{\mathbf{b}}$ we replace the diagonal matrix $\mathbf{D}$ with a block-diagonal one. We exploit **independence across clusters**:

$$\mathbb{V} \left(\sum_{g=1}^G \mathbf{x}_g' \boldsymbol{\varepsilon}_g \mid \mathbf{x} \right) = \sum_{g=1}^G \mathbb{V} (\mathbf{x}_g' \boldsymbol{\varepsilon}_g \mid \mathbf{x}_g) = \sum_{g=1}^G \mathbf{x}_g' \mathbb{E} [\boldsymbol{\varepsilon}_g \boldsymbol{\varepsilon}_g' \mid \mathbf{x}_g] \mathbf{x}_g = \sum_{g=1}^G \mathbf{x}_g' \Sigma_g \mathbf{x}_g \equiv \boldsymbol{\Omega}_n$$

So the variance is:

$$\mathbf{V}_{\mathbf{b}} = \mathbb{V}(\mathbf{b}_{OLS} \mid \mathbf{x}) = (\mathbf{x}'\mathbf{x})^{-1} \boldsymbol{\Omega}_n (\mathbf{x}'\mathbf{x})^{-1}$$

Replace Σ_g with $\mathbf{e}_g \mathbf{e}_g'$, where $\mathbf{e}_g$ are the OLS residuals for cluster g :

$$\hat{\mathbf{V}}_{\mathbf{b}}^{\text{CR1}} = \frac{G}{G-1} \frac{n-1}{n-K} (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{x}_g' \mathbf{e}_g \mathbf{e}_g' \mathbf{x}_g \right) (\mathbf{X}'\mathbf{X})^{-1}$$
$$\hat{\mathbf{V}}_{\mathbf{b}}^{\text{CR3}} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{x}_g' \tilde{\mathbf{e}}_g \tilde{\mathbf{e}}_g' \mathbf{x}_g \right) (\mathbf{X}'\mathbf{X})^{-1}$$

- ▶ CR1 is the Stata/`fixest` default; without the leading factor it is CR0.
- ▶ Can replace $\mathbf{e}_g \rightarrow \tilde{\mathbf{e}}_g = \left(\mathbf{I} - \mathbf{x}_g (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_g' \right)^{-1} \mathbf{e}_g$ for leave-one-cluster-out residuals like *HC3* (these are called *CR3*).

```
feols(y~ x1 + x2, data=df, vcov=~group_id )  
feols(y~ x1 + x2, data=df, vcov=~group_id+time_id)
```

How should I cluster my standard errors?

- ▶ Heck if I know.
- ▶ This is very problem specific
- ▶ It matters a lot → standard errors can get orders of magnitude larger.
- ▶ Do you believe across group independence or not? [this is the only thing that matters]
- ▶ If you include **fixed effects** probably you need at least clustering at that level.

Newey West Standard Errors (HAC)

- ▶ In serially correlated data we need to account for $\mathbb{C}(\varepsilon_t, \varepsilon_{t-l}) \neq 0$ for lags $l \geq 1$.
- ▶ Clustering is one solution, but we may end up throwing away all of our data.
- ▶ Instead we could estimate the serial correlation.
- ▶ May also want standard errors that are **heteroskedasticity AND autocorrelation consistent** (HAC).
- ▶ Have to select a number of lags L

$$\widehat{\Omega}_{n,L}^{HAC} = \sum_{t=1}^T e_t^2 \mathbf{x}_t \mathbf{x}_t' + \sum_{l=1}^L \sum_{t=l+1}^T w_l e_t e_{t-l} (\mathbf{x}_t \mathbf{x}_{t-l}' + \mathbf{x}_{t-l} \mathbf{x}_t')$$
$$w_l = 1 - \frac{l}{L+1}$$

You will see these constantly in accounting and finance, where serially correlated outcomes are the norm.

What about β ?

- ▶ All of the estimates above should produce **identical** point estimates
- ▶ We have just been talking about adjusting **standard errors**
- ▶ Should the presence of heteroskedasticity change our estimates $\mathbf{b}_{OLS}$ as well?

WLS, GLS, FGLS: why you will rarely run them

- ▶ In principle, if you *knew* the error variances you could reweight observations ($w_i \propto 1/\sigma_i^2$) and beat OLS on efficiency:

$$\mathbf{b}_{WLS} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}, \quad \mathbf{W} = \text{diag}(w_1, \dots, w_n) \quad (\text{GLS: } \mathbf{W} = \mathbb{V}(\boldsymbol{\varepsilon}|\mathbf{X})^{-1}, \text{ not necessarily diagonal})$$

- ▶ In practice σ_i^2 is unknown; estimating it (FGLS/IRLS) imports a model of the variance into your estimate of β , and misspecifying that model can do real damage.
- ▶ The modern division of labor: **OLS for the point estimate, robust/clustered/HAC standard errors for inference**. That is what every applied paper you read this semester does.
- ▶ The one live exception: **sampling weights** from survey data (your sample is 75% women but the population isn't) — that's a weighting story about representativeness, not about efficiency.

- ▶ Another approach to standard errors entirely: **simulate the sampling distribution** instead of deriving it.
- ▶ Resample your data with replacement, re-estimate, repeat 10,000 times, and take the standard deviation of the estimates.
- ▶ Works for quantities with no textbook formula at all (ratios of coefficients, model predictions, whole counterfactuals).
- ▶ Full treatment — bias correction, confidence intervals, when it fails, variants for clustered data — in part (b) of today's lecture.